

Scalability & Future Plan

AI-ML & GEN-AI Track | Project Phase Documentation

Date	30 September 2026	Team ID	SWTID-2026-8540
Project Name	AI BlogNest API	Track	AI / ML / Generative AI
Team Size	5	Maximum Marks	1 Mark
Team Leader	Pravin kumar .P	Members	Abdul Rahaman .A; Dhamotharan .G; Maharaja .C; SUBHASH S

Scalability & Future Plan

The plan below extends the architecture described in the reference documentation without changing the core project scope.

S.No	Current Limitation	Impact	Priority
1	AI usage and model dependency	Generation can depend on external service availability / quota	High
2	Database growth	More users/posts require indexing and query optimization	High
3	Monitoring depth	Basic logging may not be enough for larger deployments	Medium

Scalability Plan

Aspect	Current State	Proposed Upgrade / Solution
User Load	Single modular API service	Add horizontal scaling, load balancing and stronger caching where appropriate.
Data Storage	MongoDB with Mongoose	Add indexes, archival strategy and optimized queries.
Performance	Synchronous request handling	Profile hotspots and move long-running tasks to asynchronous workers if needed.
Security	JWT , RBAC, validation, sanitization and rate limiting	Add centralized secrets management, audit trails and stronger monitoring.

Future Roadmap

Phase	Planned Enhancement	Target Timeline	Expected Impact
Phase 2	Advanced analytics and richer content search	Post-MVP	Better content visibility and operational metrics
Phase 3	Scheduled publishing and background AI jobs	Future release	Improved workflow automation
Phase 4	Collaborative editing and extensible AI tools	Future release	Higher team productivity and extensibility